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1  # Note 14: K-Means Algorithm
2  # =====
3
4  # Import Libraries
5  import matplotlib.pyplot as plt
6  from sklearn.cluster import KMeans
7  from sklearn.metrics import silhouette_score
8
9  # Define Variables
10 X = [4, 5, 10, 4, 3, 11, 14, 6, 10, 12]
11 y = [21, 19, 24, 17, 16, 25, 24, 22, 21, 21]
12
13 # Plot the Scatter Plot
14 plt.scatter(X, y)
15 plt.grid(True)
16 plt.show()
17
18 # Set Data as Combinations
19 data = list(zip(X, y))
20 print(data)
21
22 # Define Model and Fit Data
23 kmeans = KMeans(n_clusters=5, random_state=0, n_init="auto").fit(data)
24 kmeans.fit(data)
25 print("Cluster Labels : ", kmeans.labels_)
26 print("Cluster Centers : {", kmeans.cluster_centers_[0], ", ", kmeans.cluster_centers_
27 [1], ", ")
28
29 # Plot Clustering Graph
30 plt.figure()
31 plt.scatter(X, y, c=kmeans.labels_, cmap='viridis')
32 plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1], marker='x')
33 plt.grid(True)
34 plt.show()
35 print(type(kmeans.cluster_centers_))
36
37 # Find Intertia Values
38 Inertia_Values = []
39 for key in range(1, 6):
40     kmeans = KMeans(n_clusters=key, random_state=0)
41     kmeans.fit(data)
42     Inertia_Values.append(kmeans.inertia_)
43
44 # Plot Inertia Graph
45 plt.figure()
46 plt.scatter(range(1, 6), Inertia_Values, color='red')
47 plt.plot(range(1, 6), Inertia_Values)
48 plt.show()
49
50 # Find S Score
51 sc = silhouette_score(data, kmeans.labels_, metric='euclidean')
52 print(sc)
53
54 sc_values = []
55 for key in range(2, 6):
56     kmeans = KMeans(n_clusters=key, random_state=0)
57     kmeans.fit(data)
58     sc_values.append(silhouette_score(data, kmeans.labels_))
59 print("SC Values : ", sc_values)
60
61 # Comprehensive List in Python
62 # sepal_length_min = [x / min(sepal_length) for x in sepal_length]

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